

Measuring Neural-Network Robustness Under Adversarial Perturbation

github.com/poojakira/adversarial-ml-lab

Research Brief - Poster 06

Evidence status: Refreshed against current code snapshot
8412e98d38f28c2e4b43cdcbf8269133b346ca16 and successful CI run 36783579501 on 2026-09-30.
The real CIFAR-10 robustness values are from the committed measured artifact
results/cifar10_smallcnn_real.json; the current CI run validates the code/test surface but did not rerun the expensive CIFAR-10 benchmark job.

Repository

github.com/poojakira/adversarial-ml-lab - public, default branch main.

Academic Project Title

Measuring Neural-Network Robustness Under Adversarial Perturbation

Subtitle

Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

One-Sentence Contribution

A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured **71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255**, while explicitly separating measured results from literature projections.

Method

1. Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.
2. Measure clean accuracy on the full CIFAR-10 test set.
3. Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.
4. Measure robust accuracy on the committed attack subset.
5. Emit structured evidence and CI robustness-gate logic.

Current Verified Evidence

Current-main Python 3.12 CI reports:

- **109 tests passed.**
- **32.16% statement coverage;** CI gate is 15%.
- Lint/format, security audit, and CodeQL jobs succeeded.
- The current CI robustness benchmark job is skipped on ordinary pushes, so the benchmark values below come from the committed real artifact rather than this CI run.

Committed measured CIFAR-10 artifact:

| Attack | Setting | Clean accuracy | Robust accuracy | Samples |

|---|---|---|---|---|

| FGSM | epsilon 8/255 | 71.82% | **3.32%** | 1,024 |

| PGD-20 | epsilon 8/255, alpha 2/255 | 71.82% | **0.00%** | 1,024 |

| C&W L2 | c=1.0, 100 steps | 71.82% | **4.20%** | 1,024 |

The model is a **1,117,354-parameter SmallCNN**, trained for 6 epochs on CPU. Clean accuracy is measured on all 10,000 CIFAR-10 test examples.

Correction to the Older Poster

The older poster centered results/robustbench_real.json, an ImageNet-pretrained ResNet evaluated on a tiny CIFAR-10 subset with severe domain mismatch. That artifact remains historical evidence, but it is no longer the best representation of this project's current measured result. The poster now uses results/cifar10_smallcnn_real.json.

Limitations

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10_resnet18_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.

Reproducibility

```
git clone https://github.com/poojakira/adversarial-ml-lab.git

cd adversarial-ml-lab

git checkout 8412e98d38f28c2e4b43cdcbf8269133b346ca16

python -m pip install -e ".[dev]"

pytest tests/ -q --cov=adv_lab --cov-report=term

python scripts/run_real_smallcnn_benchmark.py --epochs 6 --attack-samples 1000
```

Expected current CI evidence: **109 passed, 32.16% coverage**.